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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
19/037,617	01/27/2025	Yu Fang	SONA-0013-U01-C02	7180

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EXAMINER	
GREENE, JOSEPH L	

ART UNIT	PAPER NUMBER
2443	

NOTIFICATION DATE	DELIVERY MODE
04/28/2026	ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

uspatents@gtclawgroup.com

Office Action Summary	Application No. 19/037,617	Applicant(s) Fang et al.	
	Examiner JOSEPH L GREENE	Art Unit 2443	AIA (First Inventor to File) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

1)☑ Responsive to communication(s) filed on 01/27/2025.

☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2a)☐ This action is **FINAL.** 2b) ☑ This action is non-final.

3)☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

4)☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims*

5) ☑ Claim(s) 1-11 is/are pending in the application.

5a) Of the above claim(s) _____ is/are withdrawn from consideration.

6) ☐ Claim(s) _____ is/are allowed.

7) ☑ Claim(s) 1-11 is/are rejected.

8) ☐ Claim(s) _____ is/are objected to.

9) ☐ Claim(s) _____ are subject to restriction and/or election requirement

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to **PPHfeedback@uspto.gov.**

Application Papers

10)☐ The specification is objected to by the Examiner.

11)☐ The drawing(s) filed on _____ is/are: a)☐ accepted or b)☐ objected to by the Examiner.

Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).

Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

12)☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a)☐ All b)☐ Some** c)☐ None of the:

1.☐ Certified copies of the priority documents have been received.

2.☐ Certified copies of the priority documents have been received in Application No. _____.

3.☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

1) ☑ Notice of References Cited (PTO-892)

2) ☐ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b) Paper No(s)/Mail Date _____.

3) ☐ Interview Summary (PTO-413) Paper No(s)/Mail Date _____.

4) ☐ Other: _____.

DETAILED ACTION

Notice of Pre-AIA or AIA Status

The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

1. Claims 1 – 11 are currently pending in this application.

Double Patenting

Claims 1 of this application is patentably indistinct from claims 1 & 20 of Application No. 18/235,481. Pursuant to 37 CFR 1.78(f), when two or more applications filed by the same applicant or assignee contain patentably indistinct claims, elimination of such claims from all but one application may be required in the absence of good and sufficient reason for their retention during pendency in more than one application. Applicant is required to either cancel the patentably indistinct claims from all but one application or maintain a clear line of demarcation between the applications. See MPEP § 822.

19/037617	18/235,481
Claim 1: A system, comprising: a vehicle comprising a plurality of network zones, each network zone comprising a plurality of end points;	Claim 1: A system, comprising: a vehicle comprising a plurality of network zones comprising a first network zone and a second network zone, each network zone

	comprising a plurality of end points, wherein the first network zone and the second network zone have distinct network types comprising at least two members selected from the group consisting of: an ethernet network, controller area network (CAN), and a local interconnect network (LIN); and
a controller, comprising: a network monitoring component configured to interpret network communications associated with at least one of the network zones;	a controller, comprising: a log monitoring component configured to interpret a log corpus associated with at least one of the plurality of end points; a log analysis component configured to detect a risk event in response to the interpretation of the log corpus, wherein the log corpus is associated with a plurality of end points including a first end point on the first network zone, and a second end point on the second network zone; and

	a risk response component configured to perform a risk response operation in response to the detected risk event. Claim 20: wherein the risk response component is further configured to perform the risk response operation by determining an automated intrusion response in response to the detected risk event, and communicating the automated intrusion response to a log analysis user interface
a network intrusion detection component configured to detect an intrusion attempt in response to the network communications; and an intrusion response component configured to perform an intrusion prevention operation in response to the detected intrusion attempt.	

Accordingly, claims 2-11 are rejected, at least, based on their dependency on claim 1.

Claim Rejections - 35 USC § 102

In the event the determination of the status of the application as subject to AIA 35 U.S.C. 102 and 103 (or as subject to pre-AIA 35 U.S.C. 102 and 103) is incorrect, any correction of the statutory basis (i.e., changing from AIA to pre-AIA) for the rejection will

not be considered a new ground of rejection if the prior art relied upon, and the rationale supporting the rejection, would be the same under either status.

The following is a quotation of the appropriate paragraphs of 35 U.S.C. 102 that form the basis for the rejections under this section made in this Office action:

> A person shall be entitled to a patent unless –
>
> (a)(2) the claimed invention was described in a patent issued under section 151, or in an application for patent published or deemed published under section 122(b), in which the patent or application, as the case may be, names another inventor and was effectively filed before the effective filing date of the claimed invention.

Claim(s) 1-11 are rejected under 35 U.S.C. 102(a)(2) as being anticipated by Qiao et al. (Pre-Grant Publication No. US 2020/0274851 A1), hereinafter Qiao.

2. With respect to claim 1, Qiao taught system, comprising: a vehicle comprising a plurality of network zones, each network zone comprising a plurality of end points (0007); a controller (0029, wherein, at least, the CAN teaches the controller), comprising: a network monitoring component configured to interpret network communications associated with at least one of the network zones (0007); a network intrusion detection component configured to detect an intrusion attempt in response to the network communications (0019, the IDP); and an intrusion response component configured to perform an intrusion prevention operation in response to the detected intrusion attempt (0019, where the prevention comprises elements that make up the response).

3. As for claim 2, it is rejected on the same basis as claim 1. In addition, Qiao taught wherein the intrusion prevention operation comprises adjusting a blocked/allowed source list (0046 & 0053).

4. As for claim 3, it is rejected on the same basis as claim 1. In addition, Qiao taught wherein the intrusion prevention operation comprises adjusting a blocked/allowed target list (0059).

5. As for claim 4, it is rejected on the same basis as claim 1. In addition, Qiao taught wherein the network intrusion detection component is further configured to interpret an automated intrusion response in response to the detected intrusion attempt, and to update the operations to detect the intrusion attempt in response to the automated intrusion response (0025, where the updating can be seen in, at least, 0046).

6. As for claim 5, it is rejected on the same basis as claim 1. In addition, Qiao taught wherein the network intrusion response component is further configured to interpret an automated intrusion response in response to the detected intrusion attempt, and wherein the intrusion response component is further configured to update the intrusion prevention operation in response to the automated intrusion response (0046, wherein blocking traffic from a particular sender is updating).

7. As for claim 6, it is rejected on the same basis as claim 1. In addition, Qiao taught wherein the intrusion response component is further configured to interpret an automated intrusion response in response to the detected intrusion attempt, and to update the operations to respond to the detected intrusion attempt in response to the automated intrusion response (0046, wherein blocking traffic from a particular sender is updating).

8. As for claim 7, it is rejected on the same basis as claim 1. In addition, Qiao taught wherein the controller further comprises a vehicle cloud communication controller (0034), the vehicle cloud communication controller configured to manage communications between: 1) each one of the plurality of end points on the plurality of network zones, and 2) external devices at least selectively communicatively coupled to the vehicle (0034, where the external devices are implicitly taught by the external networks of 0008); wherein the vehicle cloud communication controller is further configured to interpret an automated intrusion response (0046); and the system further comprising at least one of: wherein the network intrusion detection component is further configured to update the operations to detect the intrusion attempt in response to the automated intrusion response; or wherein the intrusion response component is further configured to update the operations to respond to the detected intrusion attempt in response to the automated intrusion response (0024-0025, where the policies are updated).

9. As for claim 8, it is rejected on the same basis as claim 7. In addition, Qiao taught wherein the automated intrusion response comprises a workflow to be performed on the controller (figure 8).

10. As for claim 9, it is rejected on the same basis as claim 8. In addition, Qiao taught wherein the workflow comprises a workflow to perform at least one operation selected from: detecting an intrusion attempt; confirming an intrusion attempt; adjusting an intrusion notification scheme; blocking an intrusion attempt; or characterizing an intrusion attempt (0046, where this, at least, teaches the blocking limitation).

11. As for claim 10, it is rejected on the same basis as claim 7. In addition, Qiao taught wherein the automated intrusion response is received from an external device (0008, devices from the external networks).

12. As for claim 11, it is rejected on the same basis as claim 7. In addition, Qiao taught wherein the automated intrusion response is at least one of received from the intrusion response component or activated by the intrusion response component (0046, the response to the threat that was detected by the system).

Conclusion

The prior art made of record and not relied upon is considered pertinent to applicant's disclosure.

(a) Yates et al. (Pre-Grant Publication No. US 2020/0150637 A1), 0038, 0065.

(b) Overby et al. (Patent No. US 11,652,827 B2), 3:53-59, 25:60 to 26:12.

Any inquiry concerning this communication or earlier communications from the examiner should be directed to JOSEPH L GREENE whose telephone number is (571)270-3730. The examiner can normally be reached Monday - Thursday, 10:00am - 4:00pm.

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at http://www.uspto.gov/interviewpractice.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Nicholas R. Taylor can be reached at 571 272-3889. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

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Representative, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

/JOSEPH L GREENE/
Primary Examiner, Art Unit 2443